Rudrax Kumar Garwa

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.34
Jhunjhunu Academy	Class XII (CBSE)	2024	94.00%
S S Mody Vidya Vihar	Class X (CBSE)	2022	93.00%

PROJECTS

Terminal Chess Game using Alpha-Beta Pruning | Self Project

Aim: *Build a terminal-based chess game with intelligent AI using Alpha-Beta Pruning.*

- Developed a C++ chess game supporting **multiplayer** and **bot** matches with complete chess rule implementation.
- Implemented an **Alpha-Beta pruning** chess bot with adaptive strategy and a customizable three difficulty levels.
- Designed a real time console updates with board visualization, legal move validation, and move history tracking.
- Applied **object-oriented design** and robust **error handling** for modular , maintainable and extensive codebase.

Bayesian Portfolio Optimizer using C++ | Self Project

Aim: *Build a robust portfolio optimization system using Bayesian inference and quadratic programming.*

- Developed a C++ optimization framework using **Eigen** and **QuadProg++** under the practical portfolio constraints.
- Estimated the **posterior mean** and **covariance matrix** using the **Normal-Inverse-Wishart** prior on historical asset returns.
- Solved constrained Bayesian mean-variance optimization, achieving **9%** higher Sharpe ratio and a **7.7%** lower volatility.

Bank Management System | Self Project

Aim: *Build a C++ banking system for account and transaction management.*

- Developed a C++ banking system with secure authentication, fund transfers, efficient transaction management.
- Implemented role-based access control supporting **10+** core banking operations for customers and employees.
- Used **Binary Search Trees** for account management, **Stacks** for transaction history, and **Queues** for service requests.
- Designed a modular object-oriented system with robust validation for secure, reliable, and scalable banking operations.

Statistical Learning and Regression from Scratch

Aim: *Implement core statistical learning and regression algorithms from scratch to understand optimization, estimation, and numerical computation.*

- Implemented **Maximum Likelihood (MLE)**, **Maximum A posteriori (MAP)**, and covariance estimation algorithms using Python and Numpy.
- Developed a **polynomial regression** framework with both **Gradient Descent** and **Closed-Form (Normal Equation)** solutions.
- Built modular implementations for data preprocessing, matrix operations, optimization, and model evaluation without high-level ML libraries.
- Analyzed convergence, numerical stability, and model performance across different polynomial degrees using visualizations.

COMPETITIONS

- Achieved LeetCode contest rating of **1500+ (**Handle: *Rudrax*), demonstrating algorithmic problem-solving skills.
- Solved **260+** problems on GFG (Handle: *Rudrax Kumar Garwa*) and **110+** problems on LeetCode.
- Ranked among **Top 120** out of **3000+** registered IIT Hyderabad students on GFG.

SKILLS AND EXPERTISE

- **Programming Languages**: Python | C | C++ | HTML | CSS | SQL| Latex
- **Skills**: Data Structures and Algorithms | Competitive Programming | Object Oriented Programming
- **Tools and Libraries**: Git | C++ STL | GitHub | Numpy | MatPlotLib | Pytorch | Tensorflow | Pandas | Visual Studio Code | PostgreSQL | Google Colab | OverLeaf

RELEVANT COURSES

- **Programming** : Introduction to programming | Introduction to Modern AI | Intro To Classical AI | Programming for AI | Foundations of Machine Learning | Deep Learning | DBMS.
- **Mathematical** : Discrete Maths for Computer Science | Matrix Theory | Probability and Random Variable | Optimization I & II | Calculus I & II | Linear Systems and Signal Processing | Modern Physics | Differential Equations.

EXTRACURRICULARS

- Successfully qualified **Stage 1** of the AlgoUniversity Tech Fellowship (ATF) 2025, ranking among **top 8% nationwide** out of **250,000** participants.